



2807 West County Road 75
Monticello, MN 55362

June 18, 2025

L-MT-25-027
10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Monticello Nuclear Generating Plant
Docket No. 50-263
Subsequent Renewed Facility Operating License No. DPR-22

Monticello Nuclear Generating Plant Licensee Event Report 2025-003-00

Northern States Power Company, a Minnesota corporation, doing business as Xcel Energy (hereafter "NSPM"), hereby submits Licensee Event Report (LER) 50-263/2025-003-00 per 10 CFR 50.73(a)(2)(i)(B).

If you have any questions about this submittal, please contact Carrie Seipp, Senior Regulatory Engineer, at 612-330-5576.

Summary of Commitments

This letter makes no new commitments and no revisions to existing commitments.

A handwritten signature in black ink, appearing to read 'Greg D. Brown'.

Greg D. Brown
Plant Manager, Monticello Nuclear Generating Plant
Northern States Power Company – Minnesota

Enclosure

cc: Administrator, Region III, USNRC
Project Manager, Monticello, USNRC
Resident Inspector, Monticello, USNRC
State of Minnesota

ENCLOSURE

**MONTICELLO NUCLEAR GENERATING PLANT
LICENSEE EVENT REPORT 50-263/2025-003-00**

3 pages follow



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name Monticello Nuclear Generating Plant	☒ 050	2. Docket Number 263	3. Page 1 OF 3
	☐ 052		

4. Title
Condition Prohibited by Technical Specifications for not implementing setpoint change of Primary Containment Isolation Valve

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
05	12	2025	2025	- 003 -	00	06	18	2025	Facility Name	☐ 052	Docket Number

9. Operating Mode 4	10. Power Level 000
------------------------	------------------------

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact Carrie Seipp, Senior Nuclear Regulatory Engineer	Phone Number (Include area code) 612-330-5576
--	--

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
A	VB	ISV	B295	Y					

14. Supplemental Report Expected

☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)	15. Expected Submission Date	Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On May 12, 2025 at 0344 CDT, while Monticello Nuclear Generating Plant was in Mode 4 for Refueling Outage 1R32, it was discovered that the 2 Inch Inboard Torus Purge Exhaust Air Operated Valve, CV-2384, had been inoperable and potentially unable to close as required in the event of a design basis accident.

With CV-2384 inoperable, Technical Specification 3.6.1.3 "Primary Containment Isolation Valves (PCIVs)" condition A required the affected penetration be isolated with the automatic valve de-activated within 4 hours and, if condition A was not completed, condition F required the plant to be in Mode 3 in 12 hours. While CV-2384 is normally closed, since the condition of inoperability of CV-2384 was not known, no valves in this penetration had been closed and de-activated when the plant had been in Modes 1, 2, or 3. This issue is reportable under 10 CFR 50.73(a)(2)(i)(B) as an operation or condition prohibited by Technical Specifications.

The cause is that the lower bench set pressure of CV-2384 had insufficient closing force. In 2009, the Air Operated Valve Program engineer determined an updated lower bench set pressure, but no work order was performed to implement this new setpoint. The lower bench set pressure of CV-2384 was raised during Refueling Outage 1R32 and the valve operability was restored.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME Monticello Nuclear Generating Plant	☒ 050	2. DOCKET NUMBER 263	3. LER NUMBER		
	☐ 052		YEAR 2025	SEQUENTIAL NUMBER 003	REV NO. 00

NARRATIVE**EVENT DESCRIPTION**

On May 12, 2025 at 0344 CDT, while Monticello Nuclear Generating Plant (MNGP) was in Mode 4 for Refueling Outage 1R32, it was discovered that the Inboard Torus Purge Exhaust Air Operated Valve (AOV), CV-2384, [EIS System Code: VB, Manufacturer: Black, Sivals, and Bryson, 2 inch globe valve, Model Number: 70-27-1DRT] had been inoperable and potentially unable to close as required in the event of a design basis accident.

Program engineers were preparing to update to the 6th Inservice Testing Interval, which includes the initial implementation of ASME Code for Operation and Maintenance of Nuclear Power Plants (OM Code) Division 1, Mandatory Appendix IV "Preservice and Inservice Testing of Active Pneumatically Operated Valve Assemblies in Nuclear Reactor Power Plants." This requires reanalysis and update to the calculations for valves within the scope of OM Code Appendix IV. On April 21, 2025 while the plant was in Mode 5 for Refueling Outage 1R32, during review of the component calculation for CV-2384, which included the updated OM Code Appendix IV requirements, it was identified that a 2009 change to the setpoint appeared to not have been implemented. The last recorded lower bench set pressure, via a 2001 work order, was nominally 5 psig rather than the updated value of 8 psig.

A work order was added to the Refueling Outage 1R32 scope to perform a diagnostic test and setpoint adjustment for CV-2384. On May 9, 2025, the as-found lower bench set pressure was 5.26 psig. On May 10, 2025, the as-left lower bench set pressure was adjusted to 12.13 psig. On May 12, 2025, it was concluded that this valve had been inoperable and potentially unable to close as required since the as-found lower bench set pressure was less than the minimum required 5.98 psig.

Since it was concluded that CV-2384 had been inoperable, TS 3.6.1.3 Condition A should have been entered whenever the plant was in Modes 1, 2, or 3. Required Action A.1 of TS 3.6.1.3 is to "isolate the affected penetration flow path by use of at least one closed and de-activated automatic valve, closed manual valve, blind flange, or check valve with flow through the valve secured" in 4 hours. Condition F of TS 3.6.1.3 requires the plant to be in Mode 3 in 12 hours if the Required Action and associated Completion Time of Condition A is not met. Since the condition of inoperability of CV-2384 was not known prior to this discovery, no valves in this penetration had been closed and de-activated when the plant had been in Modes 1, 2, or 3. This is being reported as a condition prohibited by Technical Specifications when the plant was in Modes 1, 2, and 3 in the previous three-year reporting period.

ASSESSMENT OF SAFETY CONSEQUENCES

There were no radiological, environmental, or industrial impacts associated with this issue. The health and safety of the public and site personnel were not impacted during this issue.

CV-2384 is a normally closed 2 inch bypass valve around the normally closed 18 inch Inboard Torus Vent AOV AO-2383. Outboard of these two valves is the normally closed Torus Main Exhaust AOV AO-2896. These Primary Containment Isolation Valves are opened to vent the drywell to the Standby Gas Treatment System to maintain Primary Containment pressure or for de-inerting and inerting activities for refueling outages.

These valves automatically close upon Reactor Water Low Level and High Drywell Pressure. A review was performed for the three years prior to the discovery of this issue and it was determined that the Primary Containment Isolation function for this penetration was maintained during the period of inoperability of CV-2384 since AO-2896 was operable and able to isolate the penetration's flow path if necessary. There were no other systems, structures, or components that were inoperable or contributed to this issue.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
Monticello Nuclear Generating Plant	☒ 050	263	YEAR	SEQUENTIAL NUMBER	REV NO.
	☐ 052		2025	- 003	- 00

NARRATIVE**CAUSE OF THE EVENT**

The cause is that the lower bench set pressure of CV-2384 had insufficient closing force. In 2009, the AOV Program engineer determined an updated lower bench set pressure, but no work order was performed to implement this new setpoint.

CORRECTIVE ACTIONS

The lower bench set pressure of CV-2384 was raised during Refueling Outage 1R32 and the valve operability was restored.

The planned incorporation of OM Code Appendix IV will reduce the probability of a similar issue occurring in the future.

PREVIOUS SIMILAR EVENTS

LER 50-263/2025-002-00 was submitted on June 18, 2025 for a condition prohibited by Technical Specifications for a Primary Containment Isolation Valve in the Reactor Recirculation Sample Line. While the cause of this inoperability was different, the issue was also identified while performing an update to the valve's component calculation to the OM Code Appendix IV requirements.